For a change, this time we've decided to test motherboards that come with integrated graphics solution this month. We've chosen the few most commonly available, and the latest of chipsets to be carried in this test. With prices of computer components, including motherboards, falling steadily, there's very little that separates the older chipsets to the ones just released. The chipsets in this motherboard test will be the newly released ones from AMD, Intel and NVIDIA. A comparison test on the other lot, without graphic solutions, will be coming out really soon.

Now that we have our thoughts focused on motherboards with integrated graphics, we've obviously have to split the motherboards into two groups – motherboards for Intel processors using the LGA 775 format and the ones for AMD processors using the AM2 socket. Looking at just numbers, we received more boards for AMD processors than for Intel processors. Manufacturers of boards for Intel processors for that matter seem to make more boards for users who want to install a graphic card on their system.

Who is this motherboard test for? It could actually be for everyone! Most people would love to have a decent graphic card on their system but due to budget constraints or confusion, they compromise and go for something cheap and low-end. Months down the line, people don't want to sell off or dispose off their older card and buy something new. Onboard graphics solutions these days are actually quite powerful and you'll get an idea of what we mean when we start talking about the performance of some of the boards. To give you an idea, you should be able play games that came out before 2006 or so rather well. Even if you have plans to buy a good graphic card but can't afford it, you can settle for a motherboard from the ones we've tested. A few months down line, you can plug in a PCIe graphics card. With some of the motherboards we've received like the ones with the all new NVIDIA GeForce 9300 chipsets for Intel and the GeForce 8000 and the 7xx series for AMD, you can actually turn off the external graphic card and save power as well. You can also utilise the performance of your new graphic card along with whatever power your onboard graphics card does.

How We Tested

Testing! That must mean lots and lots of statistics and numbers — everything to do with performance. It's the case with this motherboard test, but it's not only about the numbers either. It's a little known fact to many that performance in motherboards for all desktop applications and even gaming for that matter doesn't really differ a whole lot. Typically from past generations of tested motherboards, a difference of one to four per cent is expected in applications and games alike, as long as there is a common processor or a graphics card on the system. In our case, we have motherboards with onboard graphics rendering capabilities and there is bound to be a difference of more than just one to four per cent. The test process although pretty similar to previous times, had a little more weighting assigned to the game and graphics intensive benchmarks.

We ran two fairly old games to get our scores. It wouldn't make sense to attempt running extremely GPU intensive games such as *Crysis* on the onboard graphics cards. We ran *Doom 3* and *Company of Heroes* at resolutions of 800 x 600, 1024 x 768 and 1280 x 1024. These games would give a substantial frame rate to measure, and users who do not plan on purchasing graphics cards would be running such games at lower resolutions on their 17-inch screens. Other than the game benchmarks, our usual set of regular benchmarks were used to measure performance of applications on the desktop – PC Mark 05 and the synthetic benchmark SiSoft Sandra, were also used. Weightings were given to the tests and we decided to set lower priority to SiSoft Sandra, the synthetic benchmark which has no direct relation to real world performance most of the time, but can be used to measure and compare statistics between motherboards of each platform.

Most of the other weighting was given for everything other than performance. Motherboards had to have plenty of features – be it loads of SATA and USB ports or support for multiple graphics cards. Because these are motherboards with onboard graphics solutions, it's OK to assume that most of these motherboards come with good HD video decoding capabilities.

Throughout the evaluation, we looked at the design and layout of the boards. We were looking for clean spacious layouts that allowed easy installation of the processor and the graphic card with no SATA ports located under the slot. Solid-state capacitors were also a factor, which would typically mean a better chance of a long living board. Graphics cards get hot and people know that. Now imagine something like that on your motherboard. Some of boards came with great coolers and heat dissipation units such as heatsinks and heatpipes to help keep the overactive chipset temperatures under control. Everybody loves freebies, so we also gave points to motherboards that came with additional fans or coolers that would enhance the performance or stability of the boards in any way. Freebies also included the fun stuff – games and applications to use on the system.

As for the test rig, we chose the fastest processors for either platform that we could find. We used an AMD Phenom X4 9950 for the AMD motherboards and an Intel QX9650 for the Intel ones. For the storage drives, we used two identical Seagate 7200.11 500 GB. These drives are very fast, affordable and would probably be the kind of drive someone buying a budget system with an integrated graphics solution would use. With the same idea of affordable hardware in mind, we stuck in 2 GB DDR2 of Corsair Dominator RAM on the boards. Two Corsair HX620's, which we admit are a little